THERE ARE EXACTLY NINE MODEL STRUCTURES ON THE CATEGORY OF SETS, MODULO AXIOM OF CHOICE

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Abstract. We show there are exactly nine model structures on the category of sets when one assumes the Axiom of Choice. No originality is claimed there.

Contents

1. Introduction

The title of the paper is a problem given to the author by Professor Eric Babson in a reading course on simplicial sets in 2014. We finally solved it (Theorem 6.7).

In 2010, Thomas Goodwillie stated on MathOverflow that there are nine model structures on the category **SET** of sets, and "it is a pleasant exercise to find them all ([Goo10])." We soon realised everyone is claiming the same thing, but we could not find the proof in the literature. We point out that Omar Antolín Camarena provided a nice explanation in [Cam], but we needed more detail to understand the whole picture. Hence, the creation of these notes.

This paper is organised as follows: in Section 2, we provide the definition of model structure following Mark Hovey's definition in [Hov07]. In Section 3, we recall some basic results in set theory. In particular, an equivalent form of the Axiom of Choice. In Section 4, we solve the lifting problem in **SET**. In Section 5, we give a complete list of trivial cofibrations and fibrations in **SET**. In Section 6, we prove our main theorem (Theorem 6.7) that there are exactly nine model structures on **SET** when one assumes the Axiom of Choice.

We remind readers that we are closely following Omar Antolín Camarena's proof in [Cam]. No originality is claimed in this paper.

1.1. **Acknowledgement.** Firstly, we thank Professor Eric Babson for providing this exciting problem. It took us three years to solve it completely. Secondly, we thank Omar Antolín Camarena for providing his proof [Cam]. We will never know how to solve the Lifting Problem in **SET** without it. At last but not least, we thank Professor Daniel Ramras for pointing out [Hov07, Lemma 1.1.10 on page 6]. This plays a major role in our proof of Theorem 6.7.

2. Definition of a Model Structure and Some Elementary Results

Definition 2.1 ([Hov07, Definition 1.1.1 and 1.1.2 on page 2–3]).
Let **C** be a category.

1. A map f in **C** is a **retract** of a map g in **C** if and only if there is a commutative diagram of the form

$$\begin{tikzcd} A \arrow[r] \arrow[d, "f"'] & C \arrow[r] \arrow[d, "g"'] & A \arrow[d, "f"] \\ B \arrow[r] & D \arrow[r] & B, \end{tikzcd}$$

   where the horizontal composites are identities.
2. A **functorial factorisation** is an ordered pair (α, β) of functors from the morphism category of **C** to itself, such that every morphism f in **C** can be written as

$$f = \beta(f) \circ \alpha(f).$$

   In particular, the domain (resp. codomain) of $\alpha(f)$ is the domain of f (resp. $\beta(f)$), and the codomain of $\beta(f)$ is the codomain of f.
3. Suppose $i : A \to B$ and $p : X \to Y$ are maps in **C**. Then i has the **left lifting property (LLP)** with respect to p, and p has the **right lifting property (RLP)** with respect to i, if for every commutative diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r, "g"'] \arrow[ur, dashed, "h"] & Y, \end{tikzcd}$$

   there is a lift $h : B \to Y$, such that the triangles commute.

Note that we do not assume the lifting h is unique.

Definition 2.2 (Model Structure, [Hov07, Definition 1.1.3 on page 3]). A **model structure** on a category **C** is a triple $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ of three subcategories of **C**, called weak equivalences, cofibrations, and fibrations, together with two functorial factorisations (α, β) and (γ, δ) satisfying the following properties:

MS 1 **(2-out-of-3)** If f and g are morphisms of $\mathbf{C}$ such that gf is defined and two of f, g, gf are weak equivalences, then so is the third.

MS 2 **(Retracts)** If f and g are morphisms of $\mathbf{C}$ such that f is a retract of g, and g is a weak equivalence, cofibration, or fibration, then so is f.

MS 3 **(Lifting)** Define a map to be a **trivial cofibration** (resp. **trivial fibration**) if it is both a cofibration and a weak equivalence (resp. both a fibration and a weak equivalence). Then trivial cofibrations have LLP with respect to fibrations, and cofibrations have LLP with respect to trivial fibrations.

MS 4 **(Factorisation)** For any morphism f, $\alpha(f)$ is a cofibration, $\beta(f)$ is a trivial fibration; and $\gamma(f)$ is a trivial cofibration, and $\delta(f)$ is a fibration.

If $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on $\mathbf{C}$, we denote by $\mathcal{C} \cap \mathcal{W}$ (resp. $\mathcal{F} \cap \mathcal{W}$) the trivial cofibrations (resp. trivial fibrations) of $\mathbf{C}$. Also, we write $f \in \mathcal{W}$ for f is a weak equivalence.

Lemma 2.3 ([Hov07, Lemma 1.1.10 on page 6]). Suppose the category $\mathbf{C}$ admits a model structure. Then a map is a cofibration (resp. a trivial cofibration) if and only if it has the left lifting property with respect to all trivial fibrations (resp. fibrations).

Dually, a map is a fibration (resp. a trivial fibration) if and only if it has the right lifting property with respect to all trivial cofibrations (resp. cofibrations).

Proof. The trivial cofibration part of the lemma will be proved similarly, and the fibration and trivial fibration parts follow from duality. Thus we will only prove the cofibration part of the lemma.

($\Rightarrow$): The fact that every cofibration has LLP with respect to all trivial fibrations is simply MS 3.

($\Leftarrow$): Conversely, suppose $f : X \to Z$ has LLP with respect to all trivial fibrations. Using the functorial factorisation, we can write f as the composition

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "i"'] & & Z \\ & Y \arrow[ur, "p"'] & \end{tikzcd}$$

for which i is a cofibration and p is a trivial fibration. We will show f is a retract of i, hence by MS 2, f must be a cofibration.

Observe that we have a commutative square

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & Y \arrow[d, "p"] \\ Z \arrow[r, "\mathrm{id}_Z"'] \arrow[ur, dashed, "h"] & Z. \end{tikzcd}$$

So by assumption, there exists a lifting $h : Z \to Y$ such that the triangles commute. This gives the following retract diagram

$$\begin{tikzcd} X \arrow[r, "\mathrm{id}_X"] \arrow[d, "f"'] & X \arrow[r, "\mathrm{id}_X"] \arrow[d, "i"'] & X \arrow[d, "f"] \\ Z \arrow[r, "h"'] & Y \arrow[r, "p"'] & Z, \end{tikzcd}$$

proving f is a retract of i as desired.

□

Lemma 2.3 has a powerful consequence:

Corollary 2.4. Suppose $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on a category $\mathbf{C}$. Then fibrations $\mathcal{F}$ is completely determined by lifting property with respect to trivial cofibrations $\mathcal{C} \cap \mathcal{W}$.

Dually, cofibrations $\mathcal{C}$ is completely determined by lifting property with respect to trivial fibrations $\mathcal{F} \cap \mathcal{W}$.

In other words, one only needs weak equivalences and cofibrations (or weak equivalences and fibrations) to define a model structure on a category.

Proof. Suppose $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on a category $\mathbf{C}$. We will show how to express $\mathcal{F}$ in terms of $\mathcal{C} \cap \mathcal{W}$.

If π is a morphism in $\mathbf{C}$, let $\pi^{\perp}$ be the collection of morphisms in $\mathbf{C}$ that has RLP with respect to π. Then Lemma 2.3 yields

$$\mathcal{F} = \bigcap_{\pi \in \mathcal{C} \cap \mathcal{F}} \pi^{\perp}.$$

Similarly, let ${}^{\perp}\pi$ be the collection of morphisms in $\mathbf{C}$ that has LLP with respect to π. Then Lemma 2.3 yields

$$\mathcal{C} = \bigcap_{\pi \in \mathcal{F} \cap \mathcal{W}} {}^{\perp}\pi.$$

□

Addendum 2.5. Let $\overline{\pi}$ (resp. $\underline{\pi}$) be the collection of all morphisms in a category $\mathbf{C}$ with model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ that has LLP (resp. RLP) with respect to all morphisms in $\pi^{\perp}$ (resp. ${}^{\perp}\pi$). Then

$$\mathcal{C} \cap \mathcal{W} = \bigcap_{\pi \in \mathcal{C} \cap \mathcal{W}} \overline{\pi},$$

$$\mathcal{F} \cap \mathcal{W} = \bigcap_{\pi \in \mathcal{F} \cap \mathcal{W}} \underline{\pi}.$$

On the other hand, one can obtain weak equivalences from cofibrations and fibrations. Recall MS 4 asserts that every morphism $f : X \to Z$ can be factorise as $f = pi$, for which p is a trivial fibration, and i is a cofibration. Suppose now that f is a weak equivalence, then by 2-out-of-3 we must have i is a weak equivalence as well. Thus we have the decomposition

$$\mathcal{W} = (\mathcal{F} \cap \mathcal{W}) \circ (\mathcal{C} \cap \mathcal{W}), \tag{2.1}$$

that every weak equivalence is a trivial fibration composed with a trivial cofibration. This gives another method to define a model structure on a category $\mathbf{C}$. Suppose $\mathcal{C}$ and $\mathcal{F}$ are valid candidates for cofibrations and fibrations. Then Lemma 2.3 gives $\mathcal{C} \cap \mathcal{W}$ and $\mathcal{F} \cap \mathcal{W}$, then using Equation (2.1), we get $\mathcal{W}$. Of course, one needs to check if the resulting $\mathcal{W}$ satisfies 2-out-of-3 or not.

Corollary 2.6. Suppose $\mathbf{C}$ is a category with model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F})$. Then the class of isomorphisms in $\mathbf{C}$ is contained in $\mathcal{W} \cap \mathcal{C}, \mathcal{W} \cap \mathcal{F}$.

Proof. Lemma 2.3 asserts that all isomorphisms are trivial cofibrations and trivial fibrations, hence proving the claim. □

3. Some Elementary Results in the Category of Sets

Let **SET** be the category of sets. For each set Y, we denote ϕ_Y to be the unique function $\phi_Y : \varnothing \to Y$. We recall the following facts:

Lemma 3.1. Assuming the Axiom of Choice, a function $f : X \to Y$ is surjective if and only if it has a right inverse.

Proof.
($\Rightarrow$): If $f : X \to Y$ is surjective, consider the family of non-empty sets $\left\{ f^{-1}(y) \mid y \in Y \right\}$. By the Axiom of Choice, there exists a function $g : Y \to X = \bigcup_{y \in Y} f^{-1}(y)$, such that $g(y) \in f^{-1}(y)$ for all $y \in Y$. Thus we have $(fg)(y) = y$ as desired.
($\Leftarrow$): Trivial.

□

Lemma 3.2. A function $f : X \to Y$ is injective if and only if either $X = \varnothing$, or it has a left inverse.

Proof.
($\Rightarrow$): If $f : X \to Y$ is injective with $X \neq \varnothing$, fix $x_0 \in X$, and define $g : Y \to X$ by

$$g(y) := \begin{cases} x_0 & \text{if } y \in Y - \operatorname{im}(f), \\ f^{-1}(y) & \text{if else.} \end{cases}$$

Then g is a left inverse of f.
($\Leftarrow$): Trivial.

$\square$

Lemma 3.3 (S-I Decomposition). If $f : X \to Y$ is any function, then there exists an injection k and a surjection h such that $f = hk$.

Proof. If $X = \varnothing$, then $k = \mathrm{id}_\varnothing$, and $h = \phi_Y$.

If $X \neq \varnothing$, then $Y \neq \varnothing$. We can then factorise f as

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "k"'] & & Y \\ & \mathrm{im}\,(f) \arrow[ur, "h"'] & \end{tikzcd}$$

where $k = f$, and h is the inclusion map. $\square$

Lemma 3.4 (I-S Decomposition). If $f : X \to Y$ is any function, then there exists an injection p and a surjection q such that $f = pq$.

Proof. If $X = \varnothing$, then $q = \mathrm{id}_\varnothing$, and $p = \phi_Y$.

If $X \neq \varnothing$, then we can factorise f as

$$X \xrightarrow{q} X \times Y \xrightarrow{p} Y,$$

where $q(x) := (x, f(x))$, and $q(x, y) := y$. $\square$

We will always put subscript $\neq \varnothing$ (respectively $\varnothing$) in front of a class of functions to denote a subclass of functions with non-empty (respectively, empty) domain.

4. Solution to the Lifting Problem in the Category of Sets

In this section, we answer the the following question: given a commutative square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] & Y, \end{tikzcd}$$

in **SET**, what are the criteria for the pair (π, p) to guarantee the existences of a lifting $h : B \to X$ so that the diagram

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] \arrow[ur, dashed, "H"] & Y, \end{tikzcd}$$

commutes?

Let us first fix a pair (π, p) of functions and assume such H exists for this pair, and $A \neq \varnothing \neq X$. Write $B = B_1 \cup B_2$ with

$$B_1 := \left\{ b \in B \mid \pi^{-1}(b) \neq \varnothing \right\},$$
$$B_2 := B_1^c.$$

We see, by the commutativity of the upper triangle, that g must maps the fibre $\pi^{-1}(b)$ into a single point x in X. So H is defined on B_1 by

$$H(b) := g(\pi^{-1}(b)).$$

Notice that this is equivalent to say that the set $g(\pi^{-1}(b))$ has at most one element for every elements $b \in B_1$. But the domain of H is B, so one needs to extend the definition of H to B_2 as well. This can be done by using the commutativity of the lower triangle. Let us write $Y = Y_1 \cup Y_2$, similar to the decomposition of B. If $f(b) \in Y_1$, then we know that $H(b) \in p^{-1}(f(b))$. In other word, assuming H exists, the set $p^{-1}(f(b))$ has at least one element for each $b \in B_2$.

Let us summarise what we proved so far: given a pair (π, p) with a commutative square, if a lifting H exists, then

(L1) for every $b \in B_1$, $\left|g(\pi^{-1}(b))\right| \leq 1$; and
(L2) for every $b \in B_2$, $\left|p^{-1}(f(b))\right| \geq 1$.

By negating the above two conditions, we see that if

(L3) there exists $b \in B_1$ such that $\left|g(\pi^{-1}(b))\right| > 1$; or
(L4) there exists $b \in B_2$ such that $\left|p^{-1}(f(b))\right| = 0$,

then no lifting H can exist. Equivalently, we have

Proposition 4.1.
(i). (L3) holds if and only if both π, p are not injective.
(ii). (L4) holds if and only if both π, p are not surjective.

Therefore, given a commutative square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] & Y, \end{tikzcd}$$

in **SET** with $A \neq \varnothing \neq X$, if a lifing $H : B \to X$ exists, then one of π, p is injective, and one of π, p is surjective.

Lemma 3.1 and Lemma 3.2 prove the converse of Proposition 4.1:

Proposition 4.2. Given a commutative square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] & Y, \end{tikzcd}$$

in **SET** with $A \neq \varnothing \neq X$, if one of π, p is injective, and one of π, p is surjective, then there exists a lifting $H : B \to X$.

Proof.

(1). If π is bijective, then $H = g\pi^{-1}$. Moreover, we have a similar formula when p is bijective.
(2). If π is injective but not surjective, then by Lemma 3.2 it has a left inverse $k : B \to A$, then $H = gk$. Again, we have a similar formula when p is injective but not surjective.
(3). If π is surjective but not injective, then by Lemma 3.1 it has a right inverse $j : B \to A$. Then $H = gj$, and we a similar formula when p is surjective but not injective.

□

We now study the case when one of A, X is empty.

Proposition 4.3. Suppose we have a commutative square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] & Y. \end{tikzcd}$$

in **SET**.

(1) If $A \neq \varnothing = X$, then a lifting $H : B \to X$ exists.
(2) If $A = \varnothing = X$, then a lifting $H : B \to X$ exists if and only if $B = \varnothing$.
(3) If $A = \varnothing \neq X$, then a lifting $H : B \to X$ exists if and only if either $B = \varnothing$, or p is a surjection.

Proof.

(1) This is clear, since there is no commutative square of form

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & \varnothing \arrow[d, "p"] \\ B \arrow[r, "f"'] & Y. \end{tikzcd}$$

when $A \neq \varnothing$, so a lifting exists automatically.
(2) In this case the commutative square is given by

$$\begin{tikzcd} \varnothing \arrow[r, "\mathrm{id}_\varnothing"] \arrow[d, "\phi_B"'] & \varnothing \arrow[d, "\phi_Y"] \\ B \arrow[r, "f"'] & Y. \end{tikzcd}$$

In particular, $H : B \to \varnothing$ is a function if and only if $B = \varnothing$, which is also a lifting in this case.

(3) This is just a combination of part (2) and the proof of Proposition 4.2.

□

Patching Propositions 4.1, 4.2, and 4.3 all together, we have solved the lifting problem in **SET**:

Theorem 4.4 (Solution to the Lifting Problem in the Category of Sets). Given a commutative diagram of sets

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\pi"'] & X \arrow[d, "p"] \\ B \arrow[r, "f"'] \arrow[ur, dashed, "H"] & Y, \end{tikzcd}$$

a lifting $H : B \to X$ exists if and only if

(1) either $A \neq \varnothing = X$; or

(2) at least one of of π, p is injective, and at least one of π, p is surjective.

5. All Possible Choices for Trivial Cofibrations and Fibrations in the Category of Sets

Denote the following classes of functions

Notation	Description
$\mathcal{B}$	class of all bijections
$\mathcal{I}$	class of all injections
$\mathcal{S}$	class of all surjections
$\mathcal{A}$	class of all functions

In addition, if $\mathcal{K}$ is any class of functions listed above, we denote $\mathcal{K}_\varnothing$ (resp. $\mathcal{K}_{\neq\varnothing}$) the class of functions in $\mathcal{K}$ with empty (resp. non-empty) domain. For example,

$$\mathcal{I}_\varnothing := \{\phi_X : \varnothing \to X \mid X \text{ is a set}\},$$
$$\mathcal{I}_{\neq\varnothing} := \{f : X \to Y \text{ injective} \mid X \neq \varnothing\}.$$

Lemma 5.1. Suppose $\mathcal{M}$ is a class of functions such that $\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\} \subseteq \mathcal{M}$.

Assume $\mathcal{M}$ satisfies 2-out-of-3, if $\mathcal{M}$ contains the morphism $\varnothing \to X$ for some non-empty X, then $\mathcal{M} = \mathcal{A}$.

Proof. Firstly we have

$$\mathcal{A} = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}) \cup \mathcal{I}_\varnothing.$$

Thus we just need to show $\mathcal{I}_\varnothing \subseteq \mathcal{M}$.

By assumption, we have $\mathrm{id}_\varnothing \in \mathcal{M}$. If Y is any non-empty set, then, by assumption, any function $f : X \to Y$ is in $\mathcal{M}$. Then we have the following commutative diagram

$$\begin{tikzcd} \varnothing \arrow[rr, "\phi_Y"] \arrow[dr, "\phi_X"'] & & Y \\ & X \arrow[ur, "f"'] & \end{tikzcd}$$

Since $\mathcal{M}$ satisfies 2-out-of-3, we must have $\phi_Y \in \mathcal{M}$, provinf $\mathcal{I}_\varnothing \subseteq \mathcal{M}$. $\square$

Define $\pi^\perp, \overline{\pi}$ as in the proof of Corollary 2.4. Theorem 4.4 tells us that

π	$\pi^\perp$	$\overline{\pi}$
bijection	$\mathcal{A}$	$\mathcal{B}$
non-bijective injection with $A = \varnothing$	$\mathcal{S}$	$\mathcal{I}$
non-bijective injection with $A \neq \varnothing$	$\mathcal{S} \cup \mathcal{I}_\varnothing$	$\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$
non-bijective surjection	$\mathcal{I}$	$\mathcal{S}$
neither injective nor surjective	$\mathcal{B} \cup \mathcal{I}_\varnothing$	$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$

(5.1)

Suppose $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on **SET**. Then as stated in the proof of Corollary 2.4,

$$\mathcal{F} = \bigcap_{\pi \in \mathcal{C} \cap \mathcal{W}} \pi^\perp,$$
$$\mathcal{C} \cap \mathcal{W} = \bigcap_{\pi \in \mathcal{C} \cap \mathcal{W}} \overline{\pi}.$$

We then obtain the following result

Theorem 5.2. If $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on **SET**, then the following is a complete list for all possible pair $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$:

$\mathcal{C}\cap\mathcal{W}$	$\mathcal{F}$
$\mathcal{B}$	$\mathcal{A}$
$\mathcal{A}$	$\mathcal{B}$
$\mathcal{S}$	$\mathcal{I}$
$\mathcal{I}$	$\mathcal{S}$
$\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_{\varnothing}\}$	$\mathcal{B}\cup\mathcal{I}_{\varnothing}$
$\mathcal{I}_{\neq\varnothing}\cup\{\mathrm{id}_{\varnothing}\}$	$\mathcal{S}\cup\mathcal{I}_{\varnothing}$

(5.2)

Proof. The pair $(\mathcal{C}\cap\mathcal{W},\mathcal{F})=(\mathcal{A},\mathcal{B})$ is clear. We explain how to obtain the rest.

By exhausting all possible intersections of the $\overline{\pi}$'s (resp. $\pi^{\perp}$'s) in (5.1), we get a list of candidates for the left (resp. right) column of (5.2). Use Theorem 4.4 to pair them up, and the resulting table is constructed as promised.

Finally, it is clear that every function can be factorised as $\mathcal{F}\circ(\mathcal{C}\cap\mathcal{W})$ – two of them are the S-I Decomposition 3.3 and the I-S Decomposition 3.4. So the claim holds as desired. $\square$

The proof of Theorem 5.2 can be modified to show

Addendum 5.3. If $(\mathcal{W},\mathcal{C},\mathcal{F})$ is a model structure on **SET**, then the following is a complete list for all possible pair $(\mathcal{C},\mathcal{F}\cap\mathcal{W})$:

$\mathcal{C}$	$\mathcal{F}\cap\mathcal{W}$
$\mathcal{B}$	$\mathcal{A}$
$\mathcal{A}$	$\mathcal{B}$
$\mathcal{S}$	$\mathcal{I}$
$\mathcal{I}$	$\mathcal{S}$
$\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_{\varnothing}\}$	$\mathcal{B}\cup\mathcal{I}_{\varnothing}$
$\mathcal{I}_{\neq\varnothing}\cup\{\mathrm{id}_{\varnothing}\}$	$\mathcal{S}\cup\mathcal{I}_{\varnothing}$

(5.3)

6. All Model Structures on the Category of Sets

In this section, we assume $(\mathcal{W},\mathcal{C},\mathcal{F})$ is a model structure on **SET**. The triple $(\mathcal{W},\mathcal{C},\mathcal{F})$ gives a pair $(\mathcal{C}\cap\mathcal{W},\mathcal{F})$, for which it must fall into on of the item in Theorem 5.2. Our algorithm to determine all model structures is as follow:

Step 1: start with a pair $(\mathcal{C}\cap\mathcal{W},\mathcal{F})$ in Theorem 5.2, we use Equation (2.1) to construct possible candidates for weak equivalences.

Step 2: After selecting those who satisfy 2-out-of-3, we can form the pair $(\mathcal{C},\mathcal{F}\cap\mathcal{W})$, and use Addendum 5.3 to determine the cofibrations.

Step 3: Then we will have a triple $(\mathcal{W},\mathcal{C},\mathcal{F})$, and we need to show the pair $(\mathcal{C}\cap\mathcal{W},\mathcal{F})$ is indeed the original one we start with.

Proposition 6.1. If $(\mathcal{C}\cap\mathcal{W},\mathcal{F})=(\mathcal{B},\mathcal{A})$, then $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{B},\mathcal{A})$, or $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{B},\mathcal{A},\mathcal{A})$.

Proof. We have

$$\begin{aligned}\mathcal{W} &= (\mathcal{F}\cap\mathcal{W})\circ(\mathcal{C}\cap\mathcal{W}) \\ &= (\mathcal{F}\cap\mathcal{W})\circ\mathcal{B} \\ &= \mathcal{F}\cap\mathcal{W} && \text{(Corollary 2.6)}\,.\end{aligned}$$

We look at Addendum 5.3 for all possible choices for $\mathcal{F}\cap\mathcal{W}$, and see that only $\mathcal{A}$, $\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_\varnothing\}$, and $\mathcal{B}$ satisfy 2-out-of-3.

(i) If $\mathcal{W}=\mathcal{A}$, then

$$\begin{aligned}\mathcal{B} &= \mathcal{C}\cap\mathcal{W} \\ &= \mathcal{C}\cap\mathcal{A} \\ &= \mathcal{C}.\end{aligned}$$

So we get the model structure $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{B},\mathcal{A})$.

(ii) If $\mathcal{W}=\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_\varnothing\}$, then we have $(\mathcal{C},\mathcal{F}\cap\mathcal{W})=(\mathcal{C},\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_\varnothing\})$, which by Theorem 5.2 we must have $\mathcal{C}=\mathcal{B}\cup\mathcal{I}_\varnothing$. But then we get $\mathcal{C}\cap\mathcal{W}\neq\mathcal{B}$, which is absurd.

Therefore, $\mathcal{W}\neq\mathcal{A}_{\neq\varnothing}\cup\{\mathrm{id}_\varnothing\}$.

(iii) If $\mathcal{W}=\mathcal{B}$, then $(\mathcal{C},\mathcal{F}\cap\mathcal{W})=(\mathcal{C},\mathcal{B})$. By Theorem 5.2, we must have $\mathcal{C}=\mathcal{A}$, and we get the model structure $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{B},\mathcal{A},\mathcal{A})$.

□

Proposition 6.2. If $(\mathcal{C}\cap\mathcal{W},\mathcal{F})=(\mathcal{A},\mathcal{B})$, then $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{A},\mathcal{B})$.

Proof. In this case, we have $\mathcal{C}=\mathcal{W}=\mathcal{A}$, and $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{A},\mathcal{B})$ is clearly a valid model structure. □

Proposition 6.3. If $(\mathcal{C}\cap\mathcal{W},\mathcal{F})=(\mathcal{I},\mathcal{S})$, then $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{I},\mathcal{S})$.

Proof. If $f:X\to Y$ is a surjection, then by Lemma 3.1 it has a right inverse $g:Y\to X$. Since g and id_Y are weak equivalences, by 2-out-of-3 we have f is a weak equivalence as well. This shows all fibrations are weak equivalences:

$$\begin{aligned}\mathcal{F}\cap\mathcal{W} &= \mathcal{F} \\ &= \mathcal{S}.\end{aligned}$$

Thus

$$\begin{aligned}\mathcal{W} &= (\mathcal{F}\cap\mathcal{W})\circ(\mathcal{C}\cap\mathcal{W}) \\ &= \mathcal{S}\circ\mathcal{I} \\ &= \mathcal{A} && \text{(Lemma 3.3)}\,.\end{aligned}$$

This also gives $\mathcal{C}=\mathcal{I}$, and we get the model structure $(\mathcal{W},\mathcal{C},\mathcal{F})=(\mathcal{A},\mathcal{I},\mathcal{S})$. □

Proposition 6.4. If $(\mathcal{C} \cap \mathcal{W}, \mathcal{F}) = (\mathcal{S}, \mathcal{I})$, then $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{S}, \mathcal{I})$.

Proof. Suppose $f : X \to Y$ is an injection with non-empty domain. Then there exists a surjection $g : Y \to X$ such that $gf = \mathrm{id}_X$. By 2-out-of-3, we then have f is a weak equivalence, thus $\mathcal{I}_{\neq\varnothing} \subseteq \mathcal{W}$, and therefore $\mathcal{I}_{\neq\varnothing} \subseteq \mathcal{F} \cap \mathcal{W}$.

Next, because $\mathcal{W} = (\mathcal{F} \cap \mathcal{W}) \circ (\mathcal{C} \cap \mathcal{W})$, Lemma 3.4 asserts that $\mathcal{A}_{\neq\varnothing} \subseteq \mathcal{W}$. Because $\mathcal{B} \subseteq \mathcal{W}$ as well (by Lemma 2.6), we must have $\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\} \subseteq \mathcal{W} \subseteq \mathcal{A}$. Lemma 5.1 then implies either $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, or $\mathcal{W} = \mathcal{A}$.

(i) If $\mathcal{W} = \mathcal{A}$, then we get $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{S}, \mathcal{I})$.
(ii) If $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, then $(\mathcal{C}, \mathcal{F} \cap \mathcal{W}) = (\mathcal{C}, \mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\})$. But this pair is not on (5.3), so it does not produce a model structure.

□

Proposition 6.5. If $(\mathcal{C} \cap \mathcal{W}, \mathcal{F}) = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{B} \cup \mathcal{I}_\varnothing)$, then $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{A}, \mathcal{B} \cup \mathcal{I}_\varnothing)$, or $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{B} \cup \mathcal{I}_\varnothing)$.

Proof. Firstly we have

$$\mathcal{W} = (\mathcal{F} \cap \mathcal{W}) \circ (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}).$$

Since all identity maps are trivial fibrations, we have $\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\} \subseteq \mathcal{W}$. Lemma 5.1 then implies either $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, or $\mathcal{W} = \mathcal{A}$.

(i) If $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, then $\mathcal{F} \cap \mathcal{W} = \mathcal{B}$, so Addendum 5.3 gives $\mathcal{C} = \mathcal{A}$, and we get a model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{A}, \mathcal{B} \cup \mathcal{I}_\varnothing)$.
(ii) If $\mathcal{W} = \mathcal{A}$, then $\mathcal{F} \cap \mathcal{W} = \mathcal{B} \cup \mathcal{I}_\varnothing$. So Addendum 5.3 gives $\mathcal{C} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, and we get the model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{B} \cup \mathcal{I}_\varnothing)$.

□

Proposition 6.6. If $(\mathcal{C} \cap \mathcal{W}, \mathcal{F}) = (\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{S} \cup \mathcal{I}_\varnothing)$, then $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{S} \cup \mathcal{I}_\varnothing)$, or $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{I}, \mathcal{S} \cup \mathcal{I}_\varnothing)$.

Proof. We have

$$\begin{aligned}\mathcal{W} &= (\mathcal{F} \cap \mathcal{W}) \circ (\mathcal{C} \cap \mathcal{W}) \\ &= (\mathcal{F} \cap \mathcal{W}) \circ (\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}).\end{aligned}$$

Using the same argument in the proof of Proposition 6.3, we have $\mathcal{S} \subseteq \mathcal{W}$. Secdonly, S-I Decomposition 3.3 gives

$$\mathcal{A}_{\neq\varnothing} = \mathcal{S} \circ \mathcal{I}_\varnothing.$$

Since $\mathcal{I}_{\neq\varnothing} \subseteq \mathcal{W}$, 2-out-of-3 gives $\mathcal{A}_\varnothing \subseteq \mathcal{W}$. As a result, either $\mathcal{W} = \mathcal{A}$, or $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$ (Lemma 5.1).

(i) If $\mathcal{W} = \mathcal{A}$, then Addendum 5.3 gives $\mathcal{C} = \mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, and we get a model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}, \mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{S} \cup \mathcal{I}_\varnothing))$.

(ii) If $\mathcal{W} = \mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$, then we compute

$$\begin{aligned}\mathcal{F} \cap \mathcal{W} &= (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}) \cap (\mathcal{S} \cup \mathcal{I}_\varnothing)) \\ &= \mathcal{S}.\end{aligned}$$

Addendum 5.3 then gives $\mathcal{C} = \mathcal{I}$, and we get a model structure $(\mathcal{W}, \mathcal{C}, \mathcal{F}) = (\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}, \mathcal{I}, \mathcal{S} \cup \mathcal{I}_\varnothing)$.

□

We comment that we could start from the pair $(\mathcal{C}, \mathcal{F} \cap \mathcal{W})$ instead. A slight modification of our proof would give the same model structures.

Finally, we have the complete list of model structures on **SET**.

Theorem 6.7. Assuming the Axiom of Choice, there are exactly nine model structures on **SET**.

$\mathcal{W}$	$\mathcal{C}$	$\mathcal{F}$	$\mathcal{C} \cap \mathcal{W}$	$\mathcal{F} \cap \mathcal{W}$
$\mathcal{A}$	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A}$	$\mathcal{B}$
$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{A}$
$\mathcal{A}$	$\mathcal{I}$	$\mathcal{S}$	$\mathcal{I}$	$\mathcal{S}$
$\mathcal{A}$	$\mathcal{S}$	$\mathcal{I}$	$\mathcal{S}$	$\mathcal{I}$
$\mathcal{A}$	$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{B} \cup \mathcal{I}_\varnothing$	$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{B} \cup \mathcal{I}_\varnothing$
$\mathcal{A}$	$\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{S} \cup \mathcal{I}_\varnothing$	$\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{S} \cup \mathcal{I}_\varnothing$
$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{A}$	$\mathcal{B} \cup \mathcal{I}_\varnothing$	$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{B}$
$\mathcal{A}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{I}$	$\mathcal{S} \cup \mathcal{I}_\varnothing$	$\mathcal{I}_{\neq\varnothing} \cup \{\mathrm{id}_\varnothing\}$	$\mathcal{S}$
$\mathcal{B}$	$\mathcal{A}$	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{B}$

(6.1)

Proof. If $(\mathcal{W}, \mathcal{C}, \mathcal{F})$ is a model structure on **SET**, then $(\mathcal{C} \cap \mathcal{W}, \mathcal{F})$ falls into one of the pair in (5.2). Then Proposition 6.1– 6.6 gives one of the nine in (6.1). □

Index

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